

Classification: Logistic Regression

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DEFINITION

Classification is a supervised learning task where the goal is to predict discrete class labels for given input features. Logistic regression is a classification method that models the probability of a class label as a function of the input features.

LOGISTIC REGRESSION

Logistic regression predicts a categorical dependent variable based on one or more independent variables. It estimates the probability of a binary outcome using the logistic function.

LOGISTIC CURVE

The logistic curve is an S-shaped curve that represents the relationship between the independent variables and the probability of the binary outcome. It is defined by the logistic function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where z is the linear combination of the independent variables and their coefficients.